CCL-1: Finite Encodings and Shrinking Cylinders over the Gaussian Rationals

Anonymous

August 14, 2026

Abstract

Continued fractions and Gosper's continued logarithms repeatedly remove a coarse integral or dyadic part and then apply a homographic inverse map. We give two analogous finite algorithms over the Gaussian rationals. The first is a coordinate-floor complex continued fraction

$$z_n = t_n + \frac{1+i}{z_{n+1}}, \qquad t_n \in \mathbb{Z}[i].$$

Its termination proof is reduced to an explicit finite eight-digit contraction certificate; no unrecorded computer-algebra calculation is used. The second algorithm, the complex continued logarithm (CCL), emits raw binary words. It normalizes the Gaussian norm by powers of $1+i$, emits three bits selecting one of the eight representatives

$$\pm 1, \quad \pm i, \quad \pm(1+i), \quad \pm(1-i),$$

and then inverts the contracted remainder. A strict post-initial normalization convention gives direct denominator descent after every non-terminal sector step.

All choices are bounded searches through explicitly finite lists. Geometric regions are used only as rational predicates, not as named infinite sets. Complete CCL blocks may be abbreviated by $[k, s]$, but an arbitrary prefix can end inside the unary scale run or inside the three-bit sector code, and those unresolved bits remain part of its cylinder predicate and bit cost. The canonical truncation uses the formal projective coordinate $[1 : 0]$; no completed real or complex point and no actual infinite stream is required. A finite two-witness theorem gives the corresponding bit-prefix minimax statement without suprema over infinite cylinders.

The three angular bits are also parsed incrementally. Their first two bits are a quarter-turn frame and the last bit chooses the axial or diagonal representative; unit folds multiply only the first decoder column and preserve the represented projective point. A two-tail identity for Hermitian projective quadrance then shows where genuine numerical contraction occurs. A completed block of radial norm L contracts the Hermitian projective quadrance between every pair of compatible projective tails by the rational factor

$$q(L) = \frac{2L}{(1+L)^2} \leq \frac{1}{2}.$$

Consequently a prefix containing n completed blocks has the finite pairwise certificate $\chi^2 \leq 2^{-n}$. Individual raw bits still refine the rational predicate, but strict contraction at every bit is neither required nor true.

We also give an operational semantics for values not returned by a finite rational encoding. A boxed prefix certificate consists only of a finite word, a rational axis-aligned box, a rational witness, and finite proofs that the prefix predicate is satisfiable and that every rational realization of it lies in the box. The completed-block contraction theorem converts any fixed affine norm bound and rational witness into an explicit shrinking rational box. A progressive name answers each requested accuracy by extending the prefix and returning a nested box of arbitrarily small rational squared diagonal. Thus every observation remains finite, while successive prefixes explicitly produce better enclosures. For a finite rational name, its initial substrings already form a finite tower of prefix-minimax approximations ending at the exact value.

These finite cylinders and boxed names are the representation-level input to CCL-2. That companion paper supplies the redundant-output machinery needed to couple them persistently to RFAP zero/pole tracks and then develops arithmetic and finite calculus. Keeping that layer separate leaves the canonical encoder, parser, and contraction results here independent of any particular operation on names.

1 Introduction and constructive conventions

The ordinary continued fraction is the Euclidean algorithm presented as a homographic digit expansion. Gosper's continued logarithm replaces unbounded positive partial quotients by unary binary scale runs and a fixed subtraction branch; its arithmetic interpretation goes back to HAKMEM [2], and its dynamics and complexity have since been studied in several forms [3, 4, 5, 10]. Over the Gaussian rationals, Hurwitz's nearest-Gaussian-integer construction is the classical point of comparison [7]; modern accounts develop its geometry and dynamics [6, 8, 9].

The coordinate-floor CCF below is not proposed as a replacement for Hurwitz's theory. It is the direct Cartesian analogue of the real floor construction and provides a useful contrast with the radial–angular CCL. This revision is organized around three constructive constraints: every search is bounded by an explicit finite list, geometric information is carried by rational predicates rather than by names for point collections, and every direct observation consists of finite bit words and finite rational certificates. A value need not be returned as a completed point: it may instead be presented by a procedure that extends its current finite prefix and enclosure to any requested rational accuracy.

Two different approximation statements are kept separate. Prefix minimax optimality is an information theorem about finite tags. Numerical enclosure is supplied by a contraction theorem for Hermitian projective quadrance: although a partial unary or angular bit may only change coordinates and refine a predicate, each completed CCL block contracts the Hermitian projective quadrance between compatible tails to at most one half. On a fixed bounded affine region this gives explicit rational boxes. Every construction is consequently a finite algorithm over displayed rational data.

Every numerical object in the algorithms is rational. A Gaussian rational is a pair of rationals x, y, written $x + iy$; a Gaussian integer is a pair of integers a, b, written $a + bi$. We use $\mathbb{Q}(i)$ and $\mathbb{Z}[i]$ for these two ambient types and put

$$\mathrm{N}(x + iy) = x^2 + y^2.$$

Whenever a square, annulus, sector, or cylinder is mentioned below, it is introduced as a decidable predicate on finite rational data. Every named collection used in an argument is an explicitly finite list, such as the eight sector representatives or a bounded list of natural exponents.

No completed real or complex numbers are needed. In particular, the paper does not assign a completed value to an actual infinite stream. A finite stream is a finite list of bits, and a finite rational input has a finite output. Section 7 also permits a non-terminating refinement process, but each invocation of that process returns only a finite extension, a rational box, and finite correctness evidence. The name is this finite-precision interface, not a pre-existing completed point or a named infinite intersection. The symbol traditionally written as infinity appears only through the formal projective coordinate

$$\omega = [1 : 0].$$

It is a pair of rational coordinates, not a completed numerical value.

Natural numbers include zero. We put

$$\gamma = 1 + i, \qquad \mathrm{N}(\gamma) = 2.$$

Direct expansion gives the multiplicative identity $\mathrm{N}(zw) = \mathrm{N}(z)\,\mathrm{N}(w)$ for Gaussian rationals. In particular,

$$\mathrm{N}(\gamma^k z) = 2^k\,\mathrm{N}(z)$$

for every $k \in \mathbb{N}$.

All floor operations, comparisons, finite-list searches, and matrix products used below are computable. Every phrase of the form "obtain the exponent" refers to the output of the displayed bounded scan together with its proof. The sector table is indexed by three bits, the exceptional CCF table by two finite eight-valued indices, and both termination theorems recurse on an explicitly displayed natural-number norm.

2 Projective homographic bookkeeping

A projective rational coordinate is written $[p : q]$, where $p, q \in \mathbb{Q}(i)$ are not both zero; non-zero scalar multiples are identified. A matrix

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

acts by the finite formula

$$M[p : q] = [ap + bq : cp + dq]$$

whenever the output coordinates are not both zero. Every elementary decoder matrix below has non-zero determinant, so this condition is automatic for its finite products. For an affine input z we use $[z : 1]$. If M is a finite decoder product, its first column is non-zero and its canonical projective truncation is

$$M\omega = [a : c].$$

This formula remains meaningful when $c = 0$ and therefore avoids a division convention.

Lemma 2.1 (continued-fraction substitution)**.** *Let K be a commutative field and let $t, \mu, y \in K$. Assume that $ty + \mu$ and y are not both zero, and put*

$$A(t, \mu) = \begin{pmatrix} t & \mu \\ 1 & 0 \end{pmatrix}.$$

Then

$$A(t, \mu)[y : 1] = [ty + \mu : y].$$

If $y \neq 0$, this is the affine identity $x = t + \mu/y$. Moreover,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} A(t, \mu) = \begin{pmatrix} at + b & a\mu \\ ct + d & c\mu \end{pmatrix}.$$

Proof. Both statements are direct multiplication. □

We also use the finite matrices

$$S_\lambda = \begin{pmatrix} \lambda & 0 \\ 0 & 1 \end{pmatrix}, \qquad T_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}, \qquad I_\lambda = \begin{pmatrix} 0 & 1 \\ \lambda & 0 \end{pmatrix},$$

representing $z \mapsto \lambda z$, $z \mapsto z + s$, and $z \mapsto 1/(\lambda z)$ whenever the affine expressions are defined.

3 Bounded normalization by powers of two

The algorithms repeatedly choose a power of two surrounding a positive rational. It is important that this choice be a bounded finite search rather than an appeal to a logarithm or to unbounded minimization.

Lemma 3.1 (finite binary normalization)**.** *Let $A, B \in \mathbb{N}$ be positive.*

(a) *If $2B \leq A$, scanning the finite list $1, 2, \ldots, A$ and retaining the last k satisfying $2^k B \leq A$ returns the unique k with*

$$2^k B \leq A < 2^{k+1} B.$$

(b) *If $A < 2B$, scanning the finite list $0, 1, \ldots, B$ and returning the first m satisfying $B \leq 2^m A$ returns the unique m with*

$$B \leq 2^m A < 2B.$$

(c) *If $B < A$, scanning the finite list $0, 1, \ldots, A$ and retaining the last k satisfying $2^k B < A$ returns the unique k with*

$$2^k B < A \leq 2^{k+1} B.$$

Proof. The elementary induction $2^n \geq n+1$ supplies all required bounds. In (a), $k=1$ is admissible and $k=A$ is not, since $2^A B > A$. Maximality gives the upper inequality. In (b), $2^B A \geq B$, so the finite scan succeeds. If the first successful index is positive, the previous index fails and doubling gives the strict upper inequality; if it is zero, the assumed inequality $A < 2B$ gives it. In (c), $k=0$ is admissible and $k=A$ is not. Maximality again gives the upper inequality. Strict monotonicity of powers of two gives uniqueness in all three cases. □

For $q = A/B > 0$, parts (a) and (b) construct the exponent placing q in $[1,2)$, while part (c) constructs the exponent placing a rational $q > 1$ in $(1,2]$.

4 Two real baseline algorithms

4.1 Classical continued fractions

For $x \in \mathbb{Q}$, put $a = \lfloor x \rfloor$. If $x = a$, stop; otherwise replace x by $1/(x-a)$. Thus

$$x = a + \frac{1}{x^+},$$

and the decoder matrix update is

$$M \longmapsto M \begin{pmatrix} a & 1 \\ 1 & 0 \end{pmatrix}.$$

Proposition 4.1. *The classical algorithm terminates on every rational input. If it emits $a_0, \ldots, a_n$ and $M = A(a_0,1)\cdots A(a_n,1)$, then $M\omega = [x:1]$ for the original input x.*

Proof. Write a non-integral complete quotient as p/q with $q > 0$. The next denominator is the Euclidean remainder $p - aq$, a non-negative integer strictly smaller than q. This gives a well-founded recursion on q. The reconstruction identity follows by composing the inverse-branch equations and using the terminal equality $x_n = a_n$, equivalently $A(a_n,1)\omega = [a_n : 1]$. □

4.2 Real continued logarithms

For this subsection $x \in \mathbb{Q}$ and $x \geq 1$. If $x = 1$, stop. If $x \geq 2$, emit 1 and replace x by $x/2$. If $1 < x < 2$, emit 0 and replace x by $1/(x-1)$. The inverse branches give

$$1: \quad M \longmapsto M \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \qquad 0: \quad M \longmapsto M \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Proposition 4.2. *For $p, q \in \mathbb{N}$ with $p \geq q > 0$, the continued-logarithm algorithm terminates on p/q. Composing the inverse-branch matrices of its emitted word and applying the product to the terminal coordinate $[1:1]$ reconstructs $[p:q]$.*

Proof. Use the finite normalization of Theorem 3.1 to consolidate a run of k ones. Then

$$1 \leq \frac{p}{2^k q} < 2.$$

If equality holds on the left, the algorithm stops after the run. Otherwise the following zero bit sends

$$\frac{p}{q} \longmapsto \frac{2^k q}{p - 2^k q}.$$

The new numerator $2^k q$ is a positive integer strictly smaller than p. Grouping one unary run and its following zero into a macro-step gives well-founded recursion on the numerator. The reconstruction statement follows by composing the displayed inverse branches back from the terminal state 1. □

Formal verification:
`CCL.BaselineAlgorithms.cfEncode_reconstruct;CCL.BaselineAlgorithms.clEncode_reconstruct`.

5 A floor-rotated complex continued fraction

For $z = x + iy \in \mathbb{Q}(i)$, define the coordinate floor

$$\lfloor z \rfloor_\square = \lfloor x \rfloor + i \lfloor y \rfloor.$$

Algorithm 5.1 (CCF expansion)**.** Given $z_0 \in \mathbb{Q}(i)$, put $t_n = \lfloor z_n \rfloor_\square$. If $z_n = t_n$, stop. Otherwise put

$$z_{n+1} = \frac{\gamma}{z_n - t_n}.$$

Equivalently,

$$z_n = t_n + \frac{\gamma}{z_{n+1}},$$

so Theorem 2.1 gives

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \longmapsto \begin{pmatrix} at_n + b & a\gamma \\ ct_n + d & c\gamma \end{pmatrix}.$$

Define the square predicate

$$\mathrm{Sq}(u) \ :\Longleftrightarrow\ 0 \le \operatorname{Re} u < 1 \ \text{ and } \ 0 \le \operatorname{Im} u < 1.$$

For non-zero u satisfying $\mathrm{Sq}(u)$, put

$$F(u) = \frac{\gamma}{u} - \left\lfloor \frac{\gamma}{u} \right\rfloor_\square.$$

Then $\mathrm{Sq}(F(u))$.

Lemma 5.2 (digit cone)**.** *Let $u \neq 0$ satisfy $\mathrm{Sq}(u)$ and write*

$$\left\lfloor \frac{\gamma}{u} \right\rfloor_\square = a + bi.$$

Then

$$a \ge 1, \qquad -a - 1 \le b \le a.$$

Proof. Write $u = x + iy$. Since

$$\frac{\gamma}{u} = \frac{x+y}{x^2+y^2} + i\frac{x-y}{x^2+y^2},$$

we have $\operatorname{Re}(\gamma/u) \ge |\operatorname{Im}(\gamma/u)|$. Also

$$x + y - (x^2 + y^2) = x(1-x) + y(1-y) > 0,$$

so $\operatorname{Re}(\gamma/u) > 1$ and hence $a \ge 1$. The upper bound $b \le a$ follows from $\operatorname{Im}(\gamma/u) \le \operatorname{Re}(\gamma/u) < a + 1$. Finally,

$$-b - 1 < -\operatorname{Im}(\gamma/u) \le \operatorname{Re}(\gamma/u) < a + 1,$$

and integrality gives $b \ge -a - 1$. □

Formal verification:
`CCL.CoordinateFloor.digit_cone`.

The next lemma replaces a quantifier-elimination certificate by a finite table that translates directly into case splits and polynomial normalization.

Lemma 5.3 (three-step contraction)**.** *Let $u \neq 0$ satisfy $\mathrm{Sq}(u)$. If $v = F(u) \neq 0$ and $w = F(v)$, then*

$$\mathrm{N}(uvw) < 1.$$

Proof. Write

$$t = \left\lfloor \frac{\gamma}{u} \right\rfloor_{\square}, \qquad t' = \left\lfloor \frac{\gamma}{v} \right\rfloor_{\square},$$

so that

$$u = \frac{\gamma}{t+v}, \qquad v = \frac{\gamma}{t'+w}.$$

If $t = a + bi$, define the integer

$$\lambda(t) = \begin{cases} a^2 + b^2, & b \geq 0, \\ a^2, & b = -1, \\ a^2 + (b+1)^2, & b \leq -2. \end{cases}$$

For every q satisfying $\mathrm{Sq}(q)$, direct coordinate bounds give

$$\mathrm{N}(t+q) \geq \lambda(t).$$

Since $\mathrm{N}(w) < 2$,

$$\mathrm{N}(uvw) = \frac{4\,\mathrm{N}(w)}{\mathrm{N}(t+v)\,\mathrm{N}(t'+w)} < \frac{8}{\lambda(t)\lambda(t')}.$$

Thus the claim is immediate when $\lambda(t)\lambda(t') \geq 8$.

By Theorem 5.2, the inequality $\lambda(t) < 8$ leaves only the following finite list:

r	0	1	2	3	4	5	6	7
τ_r	$1-2i$	$1-i$	1	$1+i$	$2-2i$	$2-i$	2	$2+i$
$\lambda(\tau_r)$	2	1	1	2	5	4	4	5.

Suppose now that $t = \tau_r$, $t' = \tau_c$, and $\lambda(t)\lambda(t') < 8$. Write $w = x + iy$ and $tt' + \gamma = C + Di$. The identity

$$(t+v)(t'+w) = t(t'+w) + \gamma$$

gives

$$\mathrm{N}(uvw) = \frac{4\,\mathrm{N}(w)}{\mathrm{N}(t(t'+w)+\gamma)}.$$

It is therefore enough to prove positivity of

$$\begin{aligned} \Delta_{r,c}(x,y) &= \mathrm{N}(t(t'+w)+\gamma) - 4\,\mathrm{N}(w) \\ &= (\mathrm{N}(t) - 4)(x^2+y^2) + 2(aC+bD)x + 2(-bC+aD)y + C^2 + D^2, \end{aligned}$$

where $t = a + bi$. The following finite table records lower bounds valid whenever $0 \leq x \leq 1$ and $0 \leq y \leq 1$. A dash means that the preceding coarse estimate already applies. The cross marks a digit pair that cannot occur.

$r\backslash c$	0	1	2	3	4	5	6	7
0	0	1	5	16	–	–	–	–
1	×	0	4	10	4	6	10	16
2	0	1	4	8	5	6	9	13
3	6	4	6	10	–	–	–	–
4	–	6	10	–	–	–	–	–
5	–	5	9	–	–	–	–	–
6	–	6	10	–	–	–	–	–
7	–	9	13	–	–	–	–	–

(5.1)

This table is a finite polynomial certificate, not the output of an unrecorded decision procedure. To check it, put

$$A_r = \mathrm{N}(\tau_r) - 4.$$

When $A_r \le 0$, the separable quadratic identity

$$\begin{aligned}\Delta(x,y) &= (1-x)(1-y)\Delta(0,0) + x(1-y)\Delta(1,0)\\ &\quad + (1-x)y\Delta(0,1) + xy\Delta(1,1)\\ &\quad - A_r\big(x(1-x) + y(1-y)\big)\end{aligned}$$

shows that the smallest of the four displayed corner values is a lower bound. When $A_r > 0$, only rows $r = 0, 4, 5, 7$ occur. The remaining monotonicity checks are recorded explicitly below; each derivative bound holds on $0 \le x, y \le 1$.

r	c	$\partial_x \Delta_{r,c}$	$\partial_y \Delta_{r,c}$	minimizing corner	$\Delta_{r,c}$ there
0	0	≥ 8	≤ -12	$(0,1)$	0
0	1	≥ 8	≤ -2	$(0,1)$	1
0	2	≥ 8	≥ 6	$(0,0)$	5
0	3	≥ 8	≥ 16	$(0,0)$	16
4	1	≥ 16	≤ 0	$(0,1)$	6
4	2	≥ 16	≥ 8	$(0,0)$	10
5	1	≥ 12	≤ -2	$(0,1)$	5
5	2	≥ 12	≥ 6	$(0,0)$	9
7	1	≥ 16	≤ -6	$(0,1)$	9
7	2	≥ 16	≥ 2	$(0,0)$	13

Thus every numerical entry in (5.1) follows from an integer corner evaluation and one of the two displayed finite arguments.

The pair marked $\times$ is $(t, t') = (1-i, 1-2i)$. To exclude it, write

$$t' + w = A - iB, \qquad A = 1 + x, \qquad B = 2 - y,$$

so $B > 1$. Since

$$v = \frac{\gamma}{A - iB} = p + iq,$$

we have, with $R = A^2 + B^2$,

$$p = \frac{A - B}{R}, \qquad q = \frac{A + B}{R}, \qquad p^2 + q^2 = \frac{2}{R}.$$

If $t = 1 - i$, then

$$\operatorname{Im}\left(\frac{\gamma}{1 - i + v}\right) - 1 = \frac{q(1-q) - p(1+p)}{(1+p)^2 + (q-1)^2} = \frac{2(B-1)}{R\big((1+p)^2 + (q-1)^2\big)} > 0.$$

This contradicts $\mathrm{Sq}(u)$.

The three zero entries in (5.1) can attain equality only at $x = 0$, $y = 1$. The actual predicate $\mathrm{Sq}(w)$ has $y < 1$, so all admissible cases give $\Delta_{r,c}(x,y) > 0$. Hence $\mathrm{N}(uvw) < 1$. □

Formal verification:
`CCL.CCFContraction.three_step_contraction`.

Theorem 5.4 (finite Gaussian-rational CCFs)**.** *Algorithm 5.1 terminates on every input $z_0 \in \mathbb{Q}(i)$. If it emits $t_0, \dots, t_n$ and $M = A(t_0, \gamma) \cdots A(t_n, \gamma)$, then $M\omega = [z_0 : 1]$.*

Proof. The reconstruction statement follows by finite composition of the inverse-branch identities, with the terminal digit satisfying $A(t_n, \gamma)\omega = [t_n : 1]$. For termination, carry an uncancelled representation $z_n = P_n/Q_n$ with $P_n, Q_n \in \mathbb{Z}[i]$ and $Q_n \ne 0$. Then

$$P_{n+1} = \gamma Q_n, \qquad Q_{n+1} = P_n - t_n Q_n, \qquad \frac{Q_{n+1}}{Q_n} = z_n - t_n.$$

Execute at most three CCF clauses. If one is terminal, stop. If all three are non-terminal, Theorem 5.3 gives

$$\mathrm{N}(Q_{n+3}) < \mathrm{N}(Q_n).$$

Thus the encoder is defined by well-founded recursion on the positive natural number $\mathrm{N}(Q_n)$, with each recursive call producing at most three digits before passing to a strictly smaller measure. □

Formal verification:
`CCL.CCFEncoder.encode_reconstruct`.

6 Complex continued logarithms

6.1 The finite sector table

The ordered finite representative list is

$$\mathcal{R}_8 = (1, 1+i, i, -1+i, -1, -1-i, -i, 1-i).$$

The three-bit word uvw selects $s = (-1)^u i^v \gamma^w$. Equivalently, the following finite table defines both the sector predicate and the representative.

uvw	s	rational sector predicate for $z = x + iy$
000	1	$x \geq \frac{1}{2}, \quad -\frac{1}{2} < y < \frac{1}{2}$
001	$1+i$	$x \geq \frac{1}{2}, \quad y \geq \frac{1}{2}$
010	i	$-\frac{1}{2} < x < \frac{1}{2}, \quad y \geq \frac{1}{2}$
011	$-1+i$	$x \leq -\frac{1}{2}, \quad y \geq \frac{1}{2}$
100	-1	$x \leq -\frac{1}{2}, \quad -\frac{1}{2} < y < \frac{1}{2}$
101	$-1-i$	$x \leq -\frac{1}{2}, \quad y \leq -\frac{1}{2}$
110	$-i$	$-\frac{1}{2} < x < \frac{1}{2}, \quad y \leq -\frac{1}{2}$
111	$1-i$	$x \geq \frac{1}{2}, \quad y \leq -\frac{1}{2}$

(6.1)

The boundary convention makes sector selection a deterministic nest of rational comparisons. For s in the displayed eight-entry list, write $\mathrm{Sect}_s(z)$ for the corresponding row predicate. This is a finite family of decidable predicates.

Define three norm predicates:

$$\mathrm{Band}(z) :\Longleftrightarrow 1 \leq \mathrm{N}(z) \leq 2,$$
$$\mathrm{Entry}(z) :\Longleftrightarrow 1 \leq \mathrm{N}(z) < 2,$$
$$\mathrm{Loop}(z) :\Longleftrightarrow 1 < \mathrm{N}(z) \leq 2.$$

These are predicates, not names for collections of points.

Lemma 6.1 (sector partition)**.** *If* $\mathrm{N}(z) \geq 1$*, exactly one row of (6.1) holds.*

Proof. The inequalities $|x| < 1/2$ and $|y| < 1/2$ would imply $\mathrm{N}(z) < 1/2$. Hence at least one coordinate crosses a displayed threshold. If both do, their signs choose a diagonal row; otherwise the sign of the unique large coordinate chooses an axial row. The strict transverse inequalities in the axial rows assign every threshold line to exactly one diagonal row. □

Formal verification:
`CCL.Sectors.Sector.select_spec;CCL.Sectors.Sector.holds_unique`.

Lemma 6.2 (sector contraction)**.** *Assume* $\mathrm{Band}(z)$ *and let* s *be selected by (6.1). Then*

$$\mathrm{N}(z - s) < \frac{1}{2}.$$

Proof. Sign changes and coordinate exchange reduce the proof to $s = 1$ and $s = 1 + i$.

For $s = 1$, write $z = x + iy$ with $x \geq 1/2$ and $|y| < 1/2$. From $x^2 + y^2 \leq 2$ we get $x < 3/2$, because $x \geq 3/2$ would imply $x^2 > 2$. Hence $(x-1)^2 \leq 1/4$, while $y^2 < 1/4$.

For $s = 1 + i$, put

$$x = \frac{1}{2} + a, \qquad y = \frac{1}{2} + b, \qquad a, b \geq 0,$$

and write $p = a + b$, $q = a^2 + b^2$. The band inequalities give

$$\frac{1}{2} \leq p + q \leq \frac{3}{2}.$$

We claim $q < p$. Otherwise $p \leq q \leq p^2$. The lower band inequality excludes $p = 0$, so $p \geq 1$, whence $p+q \geq 2$, a contradiction. Therefore

$$\mathrm{N}(z - (1+i)) = \left(a - \frac{1}{2}\right)^2 + \left(b - \frac{1}{2}\right)^2 = q - p + \frac{1}{2} < \frac{1}{2}.$$

□

Formal verification:
`CCL.Sectors.Sector.select_contraction`.

6.2 Raw bitstream algorithm

Algorithm 6.3 (CCL expansion)**.** Let $z_0 \in \mathbb{Q}(i)$ be non-zero.

Initial normalization. Write $\mathrm{N}(z_0) = A/B$ with $A, B > 0$.

- If $2B \leq A$, use Theorem 3.1(a) to obtain $k \geq 1$ with

$$\mathrm{Entry}(z_0/\gamma^k),$$

  put $y_0 = z_0/\gamma^k$, and emit 1^k0.
- If $A < 2B$, use Theorem 3.1(b) to obtain $m \geq 0$ with

$$\mathrm{Entry}(\gamma^m z_0),$$

  put $y_0 = \gamma^m z_0$, and emit $0\,1^m0$.

Select s_0 by (6.1), emit its three-bit code, and put $r_0 = y_0 - s_0$. If $r_0 = 0$, stop. Otherwise put

$$z_1 = \frac{1}{\gamma r_0}.$$

Subsequent loop. By Theorem 6.2, $\mathrm{N}(z_j) > 1$. Use Theorem 3.1(c) to obtain the unique $k \geq 0$ such that

$$\mathrm{Loop}(z_j/\gamma^k).$$

Put $y_j = z_j/\gamma^k$, emit 1^k0, select s_j by (6.1), emit its three-bit code, and put $r_j = y_j - s_j$. Stop if $r_j = 0$; otherwise put

$$z_{j+1} = \frac{1}{\gamma r_j}$$

and repeat.

The initial stage uses $[1, 2)$ because its leading bit distinguishes division from multiplication. Every later stage uses $(1, 2]$. This single boundary change removes the endpoint-preservation case from the denominator argument and makes every non-terminal loop step strictly descending.

The zero input may be represented by a separate finite zero tag. It is not forced into the annular code.

6.3 Complete-block notation

The raw bitstream definition comes first because it is the representation. We now introduce notation that abbreviates only *completed* blocks.

Let $\beta(s) \in \{0,1\}^3$ be the three-bit code in (6.1). For $k \in \mathbb{N}$ and one of the eight displayed representatives, define

$$[k,s] \;:=\; 1^k 0\,\beta(s).$$

For the low initial branch, write

$$0[m,s] \;:=\; 0\,[m,s] \;=\; 0\,1^m 0\,\beta(s).$$

Thus $0[m,s]$ means one extra initial zero followed by the completed block $[m,s]$.

A complete non-zero CCL word therefore has one of the forms

$$[k_0,s_0][k_1,s_1]\cdots[k_r,s_r], \qquad k_0 \geq 1, \quad k_1,\ldots,k_r \geq 0,$$

or

$$0[m,s_0][k_1,s_1]\cdots[k_r,s_r], \qquad m \geq 0, \quad k_1,\ldots,k_r \geq 0.$$

The notation is not a replacement for raw finite words. A prefix can end after only some of the unary ones, immediately after the delimiter zero, or after one or two sector bits. Such a prefix has no lossless representation as a list of pairs $[k,s]$. Its unresolved bits nevertheless affect its cylinder predicate and its bit length. In particular,

$$|[k,s]| = k+4, \qquad |0[m,s]| = m+5.$$

6.4 Decoder matrices

For a completed regular block $[k,s]$, the encoder equations are

$$y = \frac{z}{\gamma^k}, \qquad r = y - s, \qquad z^+ = \frac{1}{\gamma r}.$$

Thus

$$z = \gamma^k \left(s + \frac{1}{\gamma z^+} \right),$$

and a Gaussian-integral decoder matrix is

$$B(k,s) = S_{\gamma^k} T_s I_\gamma = \begin{pmatrix} \gamma^{k+1}s & \gamma^k \\ \gamma & 0 \end{pmatrix}. \tag{6.2}$$

A completed block updates a prefix matrix by $M \mapsto MB(k,s)$.

For the low initial block $0[m,s]$, put

$$E_m^- = \begin{pmatrix} 1 & 0 \\ 0 & \gamma^m \end{pmatrix}.$$

Its matrix is

$$L(m,s) = E_m^- T_s I_\gamma = \begin{pmatrix} \gamma s & 1 \\ \gamma^{m+1} & 0 \end{pmatrix}. \tag{6.3}$$

The high initial branch uses $B(k,s)$ exactly as in (6.2).

6.5 Incremental angular parsing

The existing word uvw already has a useful algebraic structure. Put

$$\epsilon = (-1)^u i^v = i^{2u+v} \in (1, i, -1, -i).$$

Thus uv is the ordinary two-bit index of a quarter-turn and the last bit w chooses the axial representative ϵ or the following diagonal representative $\epsilon\gamma$. Equivalently, the integer with binary digits uvw is the cyclic octant index. No alternative angular code is needed.

Lemma 6.4 (unit covariance of the sector table)**.** *For $\epsilon \in (1, i, -1, -i)$, $h \in \{0, 1\}$, and $y \in \mathbb{Q}(i)$,*

$$\mathrm{Sect}_{\epsilon\gamma^h}(y) \quad \Longleftrightarrow \quad \mathrm{Sect}_{\gamma^h}(\epsilon^{-1}y). \tag{6.4}$$

Proof. Multiplication by a Gaussian unit rotates the coordinate inequalities by a multiple of ninety degrees. Direct substitution in the eight rows of (6.1), including their strict and non-strict boundary signs, gives (6.4). This is a finite check of four units and two canonical rows. □

Formal verification:
`CCL.SectorCovariance.holds_covariant`.

Suppose the current decoder relation is represented by

$$X = M[y : 1], \qquad M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

After reading u, put $y_1 = (-1)^u y$ and right-multiply by $S_{(-1)^u}$. In particular,

$$u = 1 : \qquad MS_{-1} = \begin{pmatrix} -a & b \\ -c & d \end{pmatrix}. \tag{6.5}$$

After reading v, put $y_2 = i^{-v} y_1$ and right-multiply by S_{i^v}. The accumulated frame is ϵ, so the temporary matrix is MS_ϵ and the remaining branch is one of the two canonical choices γ^0 or γ^1.

Let

$$C_h = T_{\gamma^h} I_\gamma \qquad (h \in \{0, 1\}).$$

The full branch satisfies the projective matrix identity

$$T_{\epsilon\gamma^h} I_\gamma \;\sim\; S_\epsilon C_h S_\epsilon, \tag{6.6}$$

where $\sim$ means equality up to multiplication by a non-zero scalar. Indeed, the right-hand matrix is exactly ϵ times the left-hand matrix. Hence a finite parser may process the angular code as two unit folds, one binary canonical branch, and one outgoing unit fold. It needs only a four-valued orientation frame and a one-bit axial–diagonal choice, rather than an unrelated eight-way matrix table.

The unit factors in (6.5)–(6.6) are exact changes of tail coordinate. Their accompanying bit predicates remove incompatible sectors, but the matrix fold itself does not yet assert numerical contraction. Section 7.4 makes this distinction precise.

6.6 Finite termination

Lemma 6.5 (inversion-stage descent)**.** *Let $y = P/Q$ with $P, Q \in \mathbb{Z}[i]$, $Q \neq 0$, and $\mathrm{Band}(y)$. Let s be the selected sector representative. If $P - sQ \neq 0$, then*

$$z^+ = \frac{Q}{\gamma(P - sQ)}$$

is an uncancelled representation of the next post-inversion state and

$$\mathrm{N}\big(\gamma(P - sQ)\big) < \mathrm{N}(Q).$$

Proof. By Theorem 6.2,

$$\mathrm{N}(P - sQ) < \frac{1}{2}\,\mathrm{N}(Q).$$

Multiplication by γ doubles the norm. □

Formal verification:
`CCL.Decoder.inversion_stage_descent`.

Theorem 6.6 (finite Gaussian-rational CCLs)**.** *Algorithm 6.3 terminates on every non-zero input $z_0 \in \mathbb{Q}(i)$. If M is the product of (6.2)–(6.3) for the complete blocks it emits, in emission order, then $M\omega = [z_0 : 1]$.*

Proof. Finite matrix composition proves the reconstruction statement: at the final exact subtraction, the unresolved post-inversion coordinate is ω, and the last block sends it to the terminal annulus value. For termination, write each annulus-stage value without cancellation as

$$y_j = \frac{P_j}{Q_j}, \qquad P_j, Q_j \in \mathbb{Z}[i], \qquad Q_j \neq 0.$$

The initial value satisfies Entry(y_0), and every later one satisfies Loop(y_j).

Assume the jth sector step is non-terminal. Before radial normalization, Theorem 6.5 gives

$$P_j^+ = Q_j, \qquad Q_j^+ = \gamma(P_j - s_j Q_j), \qquad \mathrm{N}(Q_j^+) < \mathrm{N}(Q_j).$$

The next loop normalization chooses $k \geq 0$ with

$$1 < \frac{\mathrm{N}(P_j^+)}{2^k \,\mathrm{N}(Q_j^+)} \leq 2.$$

It then sets

$$P_{j+1} = P_j^+, \qquad Q_{j+1} = \gamma^k Q_j^+.$$

Therefore

$$\mathrm{N}(Q_{j+1}) < \mathrm{N}(P_j^+) = \mathrm{N}(Q_j).$$

Thus one non-terminal sector step followed by the next finite normalization makes a recursive call with a strictly smaller positive natural denominator norm. This is a well-founded definition of the encoder. □

Formal verification:
`CCL.Encoder.encode_reconstruct`.

7 Finite-prefix cylinder semantics

7.1 Raw prefixes and parser state

A bit word is a finite list over $\{0, 1\}$. For a non-zero rational input z, let enc(z) be the finite word produced by Theorem 6.3. For a finite bit word w, define the cylinder predicate

$$\mathsf{Cyl}(w, z) \;:\Longleftrightarrow\; w \preccurlyeq \mathrm{enc}(z), \tag{7.1}$$

where $\preccurlyeq$ is the ordinary finite-list prefix relation. This is a predicate, not the name of a set. It is decidable by running the encoder only until it has emitted $|w|$ bits or has terminated.

At a radial parser phase, let x denote the current rational state before normalization. The information in an unfinished unary run is already a single rational inequality; no list of its unboundedly many possible final exponents is formed. The exact equivalences are:

phase	raw fragment just read	predicate on x
initial high	$1^r,\ r \geq 1$	$2^r \leq \mathrm{N}(x)$
initial high	$1^r 0,\ r \geq 1$	$2^r \leq \mathrm{N}(x) < 2^{r+1}$
initial low	0	$\mathrm{N}(x) < 2$
initial low	$0\,1^r,\ r \geq 1$	$2^{r-1}\,\mathrm{N}(x) < 1$
initial low	$0\,1^m 0,\ m \geq 0$	$1 \leq 2^m\,\mathrm{N}(x) < 2$
subsequent loop	$1^r,\ r \geq 1$	$2^r < \mathrm{N}(x)$
subsequent loop	$1^k 0,\ k \geq 0$	$2^k < \mathrm{N}(x) \leq 2^{k+1}$

(7.2)

These statements are immediate from the maximal or minimal bounded searches in Theorem 3.1. After a delimiter, the corresponding normalized value is x/γ^r, $\gamma^m x$, or x/γ^k, and any already-read sector bits further refine its predicate by a finite disjunction over the eight table rows.

A deterministic parser scans the finite word w from left to right. Its phase records only:

(i) whether the initial high or low header is being read;
(ii) the finite unary count accumulated in the current scale run;
(iii) zero, one, or two already-read sector bits; or
(iv) that a completed block has just updated the matrix.

For a given finite prefix the counter is bounded by $|w|$; no parser state stores future exponents or future continuations.

If the current sector-bit fragment is q of length at most two, obtain the finite candidate list Σ_q by filtering the fixed eight-entry list $\mathcal{R}_8$ for representatives whose code begins with q, without changing the order. This list has respectively eight, four, or two entries. The cylinder predicate at that point is the finite disjunction of the corresponding sector predicates. Once the third bit arrives, Σ_q has one entry and the block matrix can be applied.

This distinction is essential. The block notation $[k, s]$ summarizes completed syntax, while (7.1) uses every raw bit. For example, the ten-bit prefix

$$1110101110$$

of the code in Theorem 8.5 below contains the complete block $[3, -1 - i]$ followed by the raw fragment 110. That fragment says that the next scale exponent is exactly two but contains no sector bit. It changes the cylinder depth from seven bits to ten even though no second pair $[k, s]$ is complete.

7.2 Canonical projective truncation

The parser multiplies (6.2) or (6.3) whenever a block becomes complete. Let M_w° be the resulting product after the finite scan of w. An incomplete unary scale factor may also be accumulated, but it fixes ω, so it does not change the following coordinate. An incomplete sector code has no unique translation matrix and is therefore not applied.

Define the canonical projective record

$$\Pi(w) = \big(w,\, M_w^\circ \omega\big). \tag{7.3}$$

The first component is a finite tag and is part of the record. Distinct prefixes may have the same projective coordinate while remaining distinct records. The coordinate is not asserted to be an affine rational whose encoding begins with w; the tag places it at the corresponding formal projective boundary. Thus (7.3) is the finite tagged replacement for adjoining a projective-closure endpoint, without naming a closure collection. At a completed block boundary it is the usual matrix truncation. Before the first completed block, its coordinate is the formal point ω.

7.3 A finite bit-prefix minimax theorem

Append an external terminal marker † to a completed rational code. This marker is not a CCL bit and is not charged to the bit budget; it merely distinguishes a terminated code from a strict binary extension.

A rational cylindrical gauge is a function g assigning a positive rational $g(w)$ to every finite binary word and satisfying

$$g(wb) < g(w) \qquad (b \in \{0, 1\}).$$

The bit gauge is

$$g_2(w) = \frac{1}{2^{|w|}}.$$

For two finite well-formed tags a, b over bits with an optional final terminal marker, define

$$d_g(a, b) = \begin{cases} 0, & a = b, \\ g(\mathrm{lcp}(a, b)), & a \neq b, \end{cases} \tag{7.4}$$

where lcp is their longest common binary prefix. The usual longest-common-prefix argument gives the strong triangle inequality, so this is an ultrametric on finite tags, not on numerical points.

Theorem 7.1 (finite two-witness minimax)**.** *Let w be a finite non-terminal prefix. Suppose there are rational witnesses z_0, z_1 such that*

$$\mathsf{Cyl}(w0, z_0), \qquad \mathsf{Cyl}(w1, z_1).$$

Put $x_b = \mathrm{enc}(z_b)\dagger$ and, for a candidate finite tag a, define the finite two-witness radius

$$R_w(a; z_0, z_1) = \max\big(d_g(a, x_0), d_g(a, x_1)\big).$$

Then:
(a) *If $w \preccurlyeq a$, then $R_w(a; z_0, z_1) = g(w)$. Moreover, for every rational z satisfying $\mathsf{Cyl}(w, z)$,*

$$d_g\big(a, \mathrm{enc}(z)\dagger\big) \le g(w).$$

(b) *If $w \not\preccurlyeq a$, then $R_w(a; z_0, z_1) > g(w)$.*
Consequently the canonical record $\Pi(w)$, whose tag is w, is minimax for this finite-prefix gauge. The optimum is not unique: every candidate tag extending w has the same radius.

Proof. Assume first that $w \preccurlyeq a$. Every compatible terminal code shares w with a, so repeated strict child monotonicity gives the displayed upper bound. If $a = w$, both witness distances equal $g(w)$. If a has a next symbol after w, at least one of x_0, x_1 has a different next symbol; this also covers the case in which a ends there with the terminal marker. Its longest common prefix with a is therefore exactly w. Hence the finite maximum is $g(w)$.

If $w \not\preccurlyeq a$, put $v = \mathrm{lcp}(a, w)$. Then v is a proper prefix of w, and both witnesses have longest common prefix v with a. Therefore their distances equal $g(v) > g(w)$. □

Formal verification:
`CCL.PrefixMinimax.finite_twoWitness_minimax`.

Corollary 7.2 (pointwise bit budget)**.** *Under the hypotheses of Theorem 7.1, a decoder that has read exactly the word w cannot have two-witness radius below $1/2^{|w|}$ in the bit gauge. The canonical record $\Pi(w)$ attains exactly $1/2^{|w|}$.*

Proof. Apply Theorem 7.1 with the bit gauge $g_2(w) = 2^{-|w|}$. □

This theorem deliberately concerns prefix information. It does not assert optimality of the numerical coordinate in (7.3), either in Euclidean distance or in Hermitian projective quadrance.

7.4 Hermitian projective quadrance and completed-block contraction

For non-zero projective coordinates define their *Hermitian projective quadrance*

$$\chi^2\big([x_0 : x_1], [y_0 : y_1]\big) = \frac{\mathrm{N}(x_0 y_1 - y_0 x_1)}{(\mathrm{N}(x_0) + \mathrm{N}(x_1))(\mathrm{N}(y_0) + \mathrm{N}(y_1))}. \tag{7.5}$$

This is the Gaussian-norm extension of projective quadrance [1]; conjugation makes the underlying coordinate form Hermitian. Writing $\overline{z}$ for Gaussian conjugation, the following rational Lagrange identity shows that $0 \le \chi^2 \le 1$:

$$\begin{aligned}(\mathrm{N}(x_0) + \mathrm{N}(x_1))(\mathrm{N}(y_0) + \mathrm{N}(y_1)) - \mathrm{N}(x_0 y_1 - y_0 x_1)\\ = \mathrm{N}\big(x_0\overline{y_0} + x_1\overline{y_1}\big) \ge 0.\end{aligned} \tag{7.6}$$

The expression is unchanged when either homogeneous coordinate pair is multiplied by a non-zero scalar.

For a matrix $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and an affine tail t, put

$$D_M(t) = \mathrm{N}(at + b) + \mathrm{N}(ct + d).$$

Proposition 7.3 (one-tail truncation identity)**.** *Let*

$$p = M\omega = [a : c], \qquad z = M[t : 1] = [at + b : ct + d],$$

where $\det M \neq 0$*. Then*

$$\chi^2(z, p) = \frac{\mathrm{N}(\det M)}{(\mathrm{N}(a) + \mathrm{N}(c))D_M(t)}. \tag{7.7}$$

Proof. The projective numerator is

$$\mathrm{N}\big((at + b)c - a(ct + d)\big) = \mathrm{N}(bc - ad).$$

The denominator is the one displayed. □

Formal verification:
`CCL.Chordal.Mat2.chordalSq_truncation`.

Proposition 7.4 (two-tail homographic identity)**.** *Let* $\det M \neq 0$ *and let* $t, u \in \mathbb{Q}(i)$*. Then*

$$\chi^2\left(M[t : 1], M[u : 1]\right) = \frac{\mathrm{N}(\det M)\,\mathrm{N}(t - u)}{D_M(t)D_M(u)}. \tag{7.8}$$

Consequently

$$\frac{\chi^2\left(M[t : 1], M[u : 1]\right)}{\chi^2\left([t : 1], [u : 1]\right)} = \frac{\mathrm{N}(\det M)(1 + \mathrm{N}(t))(1 + \mathrm{N}(u))}{D_M(t)D_M(u)} \tag{7.9}$$

when $t \neq u$*.*

Proof. The determinant of the two output columns is

$$(at + b)(cu + d) - (au + b)(ct + d) = (ad - bc)(t - u).$$

Taking Gaussian norms gives (7.8), and division by the affine formula for $\chi^2(t, u)$ gives (7.9). □

Formal verification:
`CCL.Chordal.Mat2.chordalSq_act_affine`.

Lemma 7.5 (scale and unit folds preserve the truncation expression)**.** *Let* $\lambda \in \mathbb{Q}(i)^\times$*, let* $t = \lambda t'$*, and put* $M' = MS_\lambda$*. Then*

$$M'[t' : 1] = M[t : 1], \qquad M'\omega = M\omega,$$

and

$$\begin{aligned} \mathrm{N}(\det M') &= \mathrm{N}(\lambda)\,\mathrm{N}(\det M), \\ \mathrm{N}(a\lambda) + \mathrm{N}(c\lambda) &= \mathrm{N}(\lambda)(\mathrm{N}(a) + \mathrm{N}(c)), \\ D_{M'}(t') &= D_M(t). \end{aligned} \tag{7.10}$$

Thus the value of (7.7) is exactly unchanged. If also $u = \lambda u'$*, then*

$$\chi^2\left(M'[t' : 1], M'[u' : 1]\right) = \chi^2\left(M[t : 1], M[u : 1]\right),$$

so the two-tail expression is unchanged as well.

Proof. All statements follow by multiplying M on the right by S_λ. The first-column factor in the denominator of (7.7) acquires the same norm multiplier as the determinant, while D_M and the transported point remain unchanged. For two tails, $\mathrm{N}(t' - u') = \mathrm{N}(t - u)/\,\mathrm{N}(\lambda)$ cancels the determinant multiplier in (7.8). □

Formal verification:
`CCL.ScaleFold.chordalSq_twoTail`.

A unary 1 is the case $\lambda = \gamma$: the first decoder column and the determinant norm both double. The bit $u = 1$ in the angular code is the unit case $\lambda = -1$, giving exactly (6.5); the v bit similarly uses $\lambda = i^v$. These bits strengthen the rational tail predicate, but coefficient growth alone cannot prove contraction because it is cancelled in (7.7).

For $\lambda \in \mathbb{Q}(i)^\times$ and $s \in \mathcal{R}_8$, define the completed-block map

$$F_{\lambda,s}(t) = \lambda\left(s + \frac{1}{\gamma t}\right)$$

by the projective matrix

$$A_{\lambda,s} = \begin{pmatrix} \lambda\gamma s & \lambda \\ \gamma & 0 \end{pmatrix}. \tag{7.11}$$

For $\lambda = \gamma^k$ this is $B(k,s)$ from (6.2); for $\lambda = \gamma^{-m}$, multiplication by the non-zero scalar γ^m gives $L(m,s)$ from (6.3). It sends ω to $[\lambda s : 1]$. Put

$$L = \mathrm{N}(\lambda) > 0, \qquad q(L) = \frac{2L}{(1+L)^2}. \tag{7.12}$$

Theorem 7.6 (completed-block contraction of Hermitian projective quadrance)**.** *For $j \in \{1,2\}$, let X_j be either ω or an affine coordinate $[t_j : 1]$ with $t_j \in \mathbb{Q}(i)^\times$. For every affine X_j, put*

$$y_j = s + \frac{1}{\gamma t_j}$$

and assume $\mathrm{Band}(y_j)$ *and* $\mathrm{Sect}_s(y_j)$*. Then*

$$\chi^2\big(A_{\lambda,s}X_1, A_{\lambda,s}X_2\big) \le q(L)\chi^2(X_1,X_2) \le \frac{1}{2}\chi^2(X_1,X_2). \tag{7.13}$$

If $X_1 \neq X_2$, the first inequality is strict whenever at least one tail is affine.

Proof. For an affine compatible tail, Theorem 6.2 gives

$$\mathrm{N}(y_j - s) < \frac{1}{2}, \qquad \mathrm{N}(t_j) = \frac{1}{2\,\mathrm{N}(y_j - s)} > 1. \tag{7.14}$$

Moreover

$$\mathrm{N}(\det A_{\lambda,s}) = 2L$$

and, because $\lambda\gamma t_j y_j = \lambda\gamma s t_j + \lambda$,

$$D_{A_{\lambda,s}}(t_j) = 2\,\mathrm{N}(t_j)\big(1 + L\,\mathrm{N}(y_j)\big). \tag{7.15}$$

If both tails are affine and distinct, (7.9) becomes

$$\frac{\chi^2(F_{\lambda,s}(t_1), F_{\lambda,s}(t_2))}{\chi^2(t_1,t_2)} = \frac{L}{2}\prod_{j=1}^{2}\frac{1 + 1/\mathrm{N}(t_j)}{1 + L\,\mathrm{N}(y_j)}.$$

By (7.14), each numerator factor is strictly below 2; by the band predicate, each denominator factor is at least $1 + L$. Hence the ratio is strictly below $q(L)$. Equal affine tails give zero on both sides.

If one tail is ω, symmetry reduces to $X_2 = \omega$. Since

$$A_{\lambda,s}\omega = [\lambda s : 1],$$

direct use of the affine formula gives

$$\frac{\chi^2(F_{\lambda,s}(t_1), F_{\lambda,s}(\omega))}{\chi^2(t_1,\omega)} = \frac{L}{2}\,\frac{1 + 1/\mathrm{N}(t_1)}{(1 + L\,\mathrm{N}(y_1))(1 + L\,\mathrm{N}(s))} < \frac{L}{(1+L)^2} < q(L),$$

because $\mathrm{N}(s) \geq 1$. If both tails are ω, both sides vanish. Finally,

$$(1+L)^2 - 4L = (1-L)^2 \geq 0$$

gives $q(L) \leq 1/2$. □

Formal verification:
`CCL.BlockContraction.Tail.contraction`.

A regular or high initial block with unary exponent r has $L = 2^r$. A low initial block $0[r, s]$ has $L = 2^{-r}$. In both cases the factor is

$$q_r = \frac{2^{r+1}}{(1+2^r)^2}, \qquad q_0 = \frac{1}{2}, \qquad q_r < \frac{1}{2} \quad (r > 0). \tag{7.16}$$

Thus long unary runs do improve the completed block's contraction, but only after the sector branch has been completed.

For a finite word w and a rational $Q \geq 0$, define the finite universal Hermitian-projective-quadrance bound predicate, abbreviated by

$$\mathrm{HPQBound}(w, Q) :\Longleftrightarrow \forall z, z' \in \mathbb{Q}(i)^{\times}, \quad \mathsf{Cyl}(w, z) \wedge \mathsf{Cyl}(w, z') \Longrightarrow \chi^2\big([z:1], [z':1]\big) \leq Q. \tag{7.17}$$

This is a pairwise certificate, not a supremum over a named cylinder.

Corollary 7.7 (shrinking completed-block prefixes)**.** *Let w be the raw concatenation of n well-formed completed CCL blocks from the initial state, with unary exponents $r_0, \ldots, r_{n-1}$. Then*

$$\mathrm{HPQBound}\left(w, \prod_{j<n} q_{r_j}\right), \qquad \prod_{j<n} q_{r_j} \leq 2^{-n}. \tag{7.18}$$

Proof. Take two rational inputs satisfying the same prefix predicate. Finite decoder substitution expresses each as the same outer block applied to a residual projective tail. A residual tail is affine when its rational code continues and is ω when it terminates at that block. At every shared block the corresponding two intermediate tails satisfy the hypotheses of Theorem 7.6. Iterating the theorem through the finite block list gives the product in (7.18). The unresolved final tails have Hermitian projective quadrance at most 1 by (7.6), and every factor is at most $1/2$. □

Formal verification:
`CCL.Cylinder.completedPrefix_chordBound`.

Every raw extension preserves an already proved bound: if $w \preccurlyeq v$, then (7.1) gives

$$\mathrm{HPQBound}(w, Q) \Longrightarrow \mathrm{HPQBound}(v, Q). \tag{7.19}$$

Strict improvement at every raw bit is false. For example, the initial prefix 1 means $2 \leq \mathrm{N}(z)$, while 10 means $2 \leq \mathrm{N}(z) < 4$. If $R = \mathrm{N}(z)$ and $S = \mathrm{N}(z')$ are at least 2, then

$$\mathrm{N}(z - z') \leq 2(R+S)$$

from $\mathrm{N}(z-z') + \mathrm{N}(z+z') = 2(R+S)$, and

$$\frac{2(R+S)}{(1+R)(1+S)} \leq \frac{8}{9}.$$

The last inequality follows from

$$4RS - 5R - 5S + 4 = (4S-5)(R-2) + 3(S-2) \geq 0.$$

Hence both 1 and 10 admit the universal bound $8/9$. The rational witnesses γ and $-\gamma$ satisfy both prefixes and have Hermitian projective quadrance $8/9$, so the delimiter bit produces no strict reduction at all. Partial bits refine predicates; completed blocks supply the uniform geometric contraction.

Remark 7.8 (why the expression is squared). For affine $z, w \in \mathbb{Q}(i)$,

$$\chi^2\big([z:1],[w:1]\big) = \frac{\mathrm{N}(z-w)}{(1+\mathrm{N}(z))(1+\mathrm{N}(w))},$$

and

$$\chi^2\big([z:1],\omega\big) = \frac{1}{1+\mathrm{N}(z)}.$$

We work with χ^2 itself because every term is rational on Gaussian-rational projective coordinates; taking a square root would generally leave $\mathbb{Q}$. Hermitian projective quadrance is therefore an exact algebraic error expression. Proposition 7.3 measures one canonical truncation, while Proposition 7.4 and Theorem 7.6 control the pairwise numerical width of a finite prefix. These geometric statements complement, but do not replace, prefix minimax optimality.

7.5 Finite approximation towers and progressive boxed names

A finite rational code already contains approximations to its value. The relevant substrings are its *initial* substrings, namely its prefixes. An internal substring has discarded the initial decoder state and is generally a name fragment for a different tail, not an approximation to the original input.

Proposition 7.9 (prefix tower of a finite name)**.** *Let $z \in \mathbb{Q}(i)$ be non-zero, let $v = \mathrm{enc}(z)$, put $m = |v|$, and for $0 \le n \le m$ let w_n be the first n bits of v. Then, for $0 \le n < m$,*

$$w_n \preccurlyeq w_{n+1},$$

while for every $0 \le n \le m$,

$$\mathsf{Cyl}(w_n, z).$$

If $x = v\dagger$ is the exact terminal tag, then for $n < m$,

$$d_g(w_n, x) = g(w_n).$$

The completed decoder product satisfies

$$\Pi(v) = \big(v, [z:1]\big).$$

Whenever both binary children of w_n admit rational witnesses, Theorem 7.1 says that $\Pi(w_n)$ is minimax among all records based only on that prefix. If only one child is admissible, the next bit is forced rather than informative; one first extends through the finite run of forced bits and applies the theorem at the next genuine branch.

Proof. The prefix relations and cylinder predicates follow from the definition of w_n. For $n < m$, the longest common binary prefix of w_n and $v\dagger$ is exactly w_n, which gives the distance formula. The final decoder identity is Theorem 6.6. The minimax assertion is Theorem 7.1. □

Formal verification:
`CCL.PrefixTower.distance_prefixAt`.

Thus "best approximation" here has a precise information-theoretic meaning. At every genuine branch, no method that has read only w_n can improve its worst-case prefix radius. Independently, whenever w_n ends after j completed blocks, Theorem 7.7 says that any two rational realizations of that prefix have Hermitian projective quadrance at most 2^{-j}. In particular, any rational witness carried with the prefix is a certified numerical approximant to the encoded rational. The numerical coordinate of $\Pi(w_n)$ need not change at every raw bit and is not asserted to lie inside the affine cylinder; incomplete unary or sector fragments can still refine the tag and its enclosure before enough syntax exists to apply another block matrix. At the final block of a rational name, the canonical projective coordinate is the rational itself.

We now add explicit Euclidean enclosures without introducing a completed complex point. A *rational box* is finite data

$$B = (x_-, x_+, y_-, y_+) \in \mathbb{Q}^4, \qquad x_- \le x_+, \qquad y_- \le y_+.$$

For $z = x + iy \in \mathbb{Q}(i)$, write

$$\mathrm{InBox}(B, z) \ :\Longleftrightarrow\ x_- \le x \le x_+ \text{ and } y_- \le y \le y_+.$$

Its rational squared diagonal is

$$\Delta^2(B) = (x_+ - x_-)^2 + (y_+ - y_-)^2.$$

For boxes $C = (x'_-, x'_+, y'_-, y'_+)$ and B, write $C \sqsubseteq B$ for the four finite inequalities

$$x_- \le x'_- \le x'_+ \le x_+, \qquad y_- \le y'_- \le y'_+ \le y_+.$$

This is the box-containment relation expressed entirely by rational comparisons.

Definition 7.10 (boxed prefix certificate)**.** A boxed prefix certificate is finite data (w, B, q), where w is a finite bit word, B is a rational box, and $q \in \mathbb{Q}(i)^\times$, together with proofs of

$$\mathsf{Cyl}(w, q)$$

and

$$\forall z \in \mathbb{Q}(i)^\times, \qquad \mathsf{Cyl}(w, z) \Longrightarrow \mathrm{InBox}(B, z). \tag{7.20}$$

The witness q proves finite satisfiability. It is not declared to be the value named by the certificate.

If $w \preccurlyeq v$, then

$$\mathsf{Cyl}(v, z) \Longrightarrow \mathsf{Cyl}(w, z) \tag{7.21}$$

by transitivity of finite-list prefixes. Hence extending a name prefix always refines its cylinder predicate, even though no cylinder is introduced as a named collection. A certificate (v, C, q') *refines* (w, B, q) when $w \preccurlyeq v$ and $C \sqsubseteq B$.

A completed initial normalization header already gives a coarse rational box. For a high header 1^k0, every compatible input satisfies $\mathrm{N}(z) < 2^{k+1}$ and is therefore enclosed by

$$[-2^{k+1}, 2^{k+1}] \times [-2^{k+1}, 2^{k+1}].$$

For a low header $0\,1^m0$, every compatible input satisfies $\mathrm{N}(z) < 2$ and is enclosed by $[-2, 2] \times [-2, 2]$. Every later prefix retains these bounds; additional bits permit sharper ones.

Proposition 7.11 (completed blocks give explicit rational boxes)**.** *Let w be a completed-block prefix as in Theorem 7.7, let*

$$Q_w = \prod_{j<n} q_{r_j},$$

and let $q \in \mathbb{Q}(i)^\times$ satisfy $\mathsf{Cyl}(w, q)$. Suppose $K \in \mathbb{Q}_{>0}$ satisfies

$$\mathsf{Cyl}(w, z) \Longrightarrow \mathrm{N}(z) \le K \tag{7.22}$$

for every rational realization z. Then

$$\mathsf{Cyl}(w, z) \Longrightarrow \mathrm{N}(z - q) \le (1 + K)^2 Q_w. \tag{7.23}$$

Given a positive rational ε, if

$$Q_w \le \frac{\varepsilon^2}{16(1 + K)^2}, \tag{7.24}$$

then the rational box

$$B_\varepsilon(q) = \Big(\mathrm{Re}\, q - \frac{\varepsilon}{4}, \mathrm{Re}\, q + \frac{\varepsilon}{4}, \mathrm{Im}\, q - \frac{\varepsilon}{4}, \mathrm{Im}\, q + \frac{\varepsilon}{4}\Big) \tag{7.25}$$

satisfies (7.20), and

$$\Delta^2(B_\varepsilon(q)) = \frac{\varepsilon^2}{2} \le \varepsilon^2.$$

Proof. For compatible z and q, (7.18), the affine formula for χ^2, and (7.22) give

$$\mathrm{N}(z-q) = \chi^2(z,q)(1+\mathrm{N}(z))(1+\mathrm{N}(q)) \le (1+K)^2 Q_w.$$

Under (7.24), each coordinate difference has square at most $\varepsilon^2/16$, hence absolute value at most $\varepsilon/4$. This proves membership in (7.25); its squared diagonal is a direct rational calculation.

Clearing the positive denominators in (7.24) gives the equivalent rational test $(1+K)^2 Q_w \le (\varepsilon/4)^2$. The calculation is homogeneous in the projective coordinates and therefore also covers an explicitly rescaled integral decoder realization. □

Formal verification:
`CCL.BoxTransport.completed_affine_epsilonBox_contains_of_paper_bound`.

For a completed high header 1^k0, one may take $K = 2^{k+1}$; for a completed low header $0\,1^m0$, one may take $K = 2$. Thus a deterministic refinement procedure that, on input n, returns a longer satisfiable prefix containing at least n completed blocks can be upgraded to a progressive boxed name. Given $\varepsilon > 0$, clear denominators and perform a bounded binary scan to choose n with

$$2^{-n} \le \frac{\varepsilon^2}{16(1+K)^2}.$$

Request that many blocks, use its rational witness as the centre in (7.25), and intersect with the previous certified box to retain nesting. This proves, using only finite data, that arbitrarily many completed blocks produce arbitrarily small affine enclosures on every fixed bounded initial branch.

The next lemma remains useful for partial blocks, arithmetic transducers, and sharper anisotropic boxes.

Lemma 7.12 (finite rational box transport)**.** *Let*

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad H_M(t) = \frac{at+b}{ct+d}.$$

Suppose a finite parser state expresses every compatible affine input as $H_M(t)$ for rational tails satisfying a rational predicate $P(t)$. Suppose finite evidence supplies a rational tail box T, a positive rational δ, and the implications

$$P(t) \Longrightarrow \mathrm{InBox}(T,t), \qquad P(t) \Longrightarrow \delta \le \mathrm{N}(ct+d).$$

Then finite rational interval arithmetic computes a rational box B satisfying

$$P(t) \Longrightarrow \mathrm{InBox}\big(B, H_M(t)\big).$$

If P is the finite disjunction produced by an incomplete sector code, the same construction is performed for each entry of the finite candidate list and the coordinatewise rational hull is taken.

Proof. Write $t = x + iy$, $u = at + b$, and $v = ct + d$. The real and imaginary parts of u and v are affine rational expressions in x, y, so ordinary interval addition and multiplication give rational bounds on them over T. The affine value is

$$\frac{u}{v} = \frac{u\overline{v}}{\mathrm{N}(v)}.$$

Interval arithmetic bounds the two coordinates of $u\overline{v}$. It also gives a rational upper bound for $\mathrm{N}(v)$; replacing that upper bound by its maximum with δ gives a positive rational D such that, on P,

$$0 < \delta \le \mathrm{N}(v) \le D.$$

Thus $1/\mathrm{N}(v)$ lies in the rational interval $[1/D, 1/\delta]$, and one more finite interval multiplication bounds both coordinates of u/v. A finite disjunction is handled by finitely many repetitions followed by rational minima and maxima. No extremum over a named point collection is taken. If no positive denominator-separation certificate is yet available, one retains the previous coarse box and reads more prefix bits rather than performing an uncertified division.

For completeness, an even simpler square-root-free enclosure is available. Let L_T be the endpoint-computed upper bound for $\mathrm{N}(t)$ on T, and put

$$N(M,T) = 2\big(\mathrm{N}(a)L_T + \mathrm{N}(b)\big), \qquad R(M,T,\delta) = \frac{N(M,T)}{\delta}.$$

Then $\mathrm{N}(at+b) \le N(M,T)$ and the denominator certificate gives $\mathrm{N}(H_M(t)) \le R(M,T,\delta)$. Since $R \le (R+1)^2$, both coordinates lie between $-(R+1)$ and $R+1$. This explicit symmetric rational box proves the lemma without a square root; the coordinatewise interval construction above is a sharper interchangeable construction. The finite-list hull uses only rational minima and maxima and covers every member of an incomplete sector's candidate list. □

Formal verification:
`CCL.BoxTransport.homographicBox_contains;CCL.BoxTransport.hullList_contains_of_mem`.

Given an old box B and a newly computed box C, one may replace C by the coordinatewise intersection with B. The new rational witness lies in both boxes by (7.20)–(7.21), so the intersection is certified non-empty; it is nested in B and has no larger squared diagonal.

Definition 7.13 (progressive boxed CCL name)**.** A finite boxed refinement trace is a finite list of boxed prefix certificates in which each entry refines the preceding one. A progressive boxed CCL name is specified operationally by an initial certificate and a deterministic refinement procedure which, given the current finite trace and a positive rational ε, returns a finite extension whose last box B satisfies

$$\Delta^2(B) \le \varepsilon^2. \tag{7.26}$$

Every invocation must terminate and return its finite word, finite box, rational witness, and proofs.

Proposition 7.14 (boxed names supply squared-error processes)**.** *Every progressive boxed CCL name supplies a progressive squared-error enclosure process of the kind used in CCL-2. At a certificate (w,B,q) its candidates are precisely the non-zero rational values satisfying the raw prefix predicate, its centre is the fixed rational anchor of B, and its squared radius is $\Delta^2(B)$. Raw-word extension proves persistence, and the initial box gives a uniform rational bound for all later centres.*

More explicitly, for a positive squared-error target τ, request the boxed name at

$$r(\tau) = \min\Big(1, \frac{\tau}{4}\Big).$$

Then $r(\tau) > 0$ and $r(\tau)^2 < \tau$, so the returned squared radius is strictly below τ. Consequently two progressive boxed CCL names may be passed directly to the constructive addition, subtraction, and multiplication controllers, and to division when a positive denominator-ball separation certificate is supplied.

Proof. If $0 < r \le 1$, then $r^2 \le r$. The definition also gives $r \le \tau/4$, hence

$$r^2 \le r < 4r \le \tau.$$

If (v,C,q') refines (w,B,q), (7.21) takes every candidate for v to a candidate for w, while $C \sqsubseteq B$ preserves its enclosure. Every later anchor lies in the initial box, whose explicit endpoint bound therefore bounds its quadrance. All statements concern the current finite certificates only. □

Formal verification:
`CCL.BoxedNames.Refines.witness_distance_le`.

The procedure is permitted never to emit a terminal marker. This permits it to capture a value not supplied by a finite Gaussian-rational code—the intended non-rational case—not by postulating a completed complex point, but by producing progressively longer cylinder predicates and nested rational boxes at every requested precision. Non-termination alone is not asserted to decide irrationality. No object consisting of all prefixes, all compatible values, or the intersection of all boxes is formed.

The rational witnesses also give explicit numerical approximants. If (w', B', q') refines (w, B, q), then both q and q' lie in B, because $w \preccurlyeq w'$ and (7.20)–(7.21). Therefore

$$\mathrm{N}(q' - q) \le \Delta^2(B). \tag{7.27}$$

Along a trace satisfying increasingly small instances of (7.26), the witnesses are thus a directly certified Cauchy approximation process. For a finite rational name, Theorem 7.9 is the finite version of the same process and terminates at the exact decoder value.

Remark 7.15 (interface for constructive root isolation). A constructive root-isolation argument can couple each boxed prefix certificate with exact polynomial evidence that its current box remains a valid root enclosure, and can answer a precision request by extending the CCL prefix until (7.26) holds. The returned root is then the refinement procedure itself. This is the finite-data interface needed for a constructive fundamental theorem of algebra; the root-isolation and root-existence certificates are separate from the CCL encoding developed here.

8 Exact finite examples

Example 8.1 (real examples)**.** The rational 355/113 has classical continued fraction

$$\frac{355}{113} = [3; 7, 16]$$

and continued-logarithm word

$$100010001110011100110010 1.$$

The rational 13/5 has continued-logarithm word

$$101001.$$

Example 8.2 (CCF examples)**.** Algorithm 5.1 gives

$$\frac{3+4i}{5} \rightsquigarrow (0, 1-i, 1-i, 2),$$

$$\frac{13}{5} \rightsquigarrow (2, 1+i, 1, 2+2i),$$

and

$$\frac{355}{113} \rightsquigarrow (3, 7+7i, 16).$$

Each list is finite and may be checked by exact Gaussian-rational substitution.

Example 8.3 (a short CCL code)**.** For $z = (3+4i)/5$, Algorithm 6.3 emits

$$00001100010011.$$

In completed-block notation this is

$$0[0, 1+i]\,[1, 1+i]\,[0, -1+i].$$

The annulus-stage values are

$$\frac{3+4i}{5}, \qquad \frac{1}{2}+i, \qquad -1+i,$$

so the final sector subtraction is exact.

Example 8.4 (a real-axis CCL code)**.** For 13/5, the revised boundary convention gives

$$110110110111111000110001$$

and the block expansion

$$[2, -i]\,[2, 1-i]\,[3, 1+i]\,[1, 1+i].$$

The last normalized value is $1+i$, so the stream terminates after its sector code.

Example 8.5 (the code for 355/113)**.** The exact rational 355/113 has CCL word

$$11101011101001111011111001011111111101011111111110111.$$

Its completed-block parsing is

$$[3,-1-i]\,[2,-1]\,[4,1-i]\,[2,i]\,[8,-1-i]\,[7,1-i].$$

The ten-bit prefix discussed in Section 7 is not a list of two pairs: only the first pair is complete, and the final raw fragment is 110.

Formal verification:
`CCL.Examples.threePlusFourIOverFive_raw;CCL.Examples.thirteenOverFive_raw;CCL.Examples.threeFiftyFiveOverOneThirteen_raw`.

9 Conclusion: Cartesian and polar structure

The CCF digit $t = \lfloor z \rfloor_{\square}$ records a translate of the coordinate unit square. Its two coordinates are quantized independently, so the representation is Cartesian and its digit alphabet is unbounded.

The CCL first removes a power of $\gamma = 1 + i$, whose norm is two, and then chooses one entry from the finite eight-direction table (6.1). A completed regular block is therefore summarized by

$$[k, s] = \text{unary radial exponent } k \text{ followed by finite angular representative } s.$$

This is the precise finite sense in which CCL is polar. The raw representation remains binary, and incomplete raw fragments remain semantically significant even when no pair $[k, s]$ is complete.

The two algorithms now have fully finite termination arguments. CCF uses a three-step macro-recursion certified by the finite table (5.1). CCL uses a one-block macro-recursion with strict Gaussian denominator descent. Its angular code is the ordinary binary octant index: two bits maintain a finite quarter-turn frame and one bit chooses axial versus diagonal. Unary and unit folds are exact coordinate substitutions, while every completed block contracts compatible projective tails by the explicit factor $q_r \le 1/2$.

The CCL is therefore especially well suited to finite exact computation because normalization, sector selection, incremental bit parsing, matrix decoding, pairwise Hermitian-projective-quadrance bounds, box transport, and the termination measure are all explicit calculations on rational data. Extending a raw prefix always refines its rational predicate; accumulating completed blocks gives an effective 2^{-n} Hermitian-projective-quadrance certificate, and on a bounded initial branch this produces rational boxes of any requested squared diagonal. The resulting operational name supplies the enclosure layer on which the arithmetic and finite-calculus constructions of CCL-2 build.

A Independent machine verification

The references headed "Formal verification" point to the corresponding declarations in the companion Lean development. They are not premises of the paper proofs: each mathematical argument appears in the body above. The `papers.CCL.Chapters.Paper` and `papers.CCL.Chapters.Bridge` modules collect the combined CCL-1 and CCL-2 verification surface. A warning-fatal Lean build using the repository's checked Core toolchain, the full configured-script `check-axioms.sh` audit, and an independent nanoda check verify that surface. The selected mathematical declarations have empty transitive axiom closures: neither `propext`, `Classical.choice`, `Quot.sound`, nor any other axiom appears in their proof dependencies. Private and generated proof dependencies are included; independent compiler and elaborator helpers outside those closures remain a separate audit obligation. The constructions use neither a completed scalar field nor a carrier-wide collection of values. Build and checking commands are given in the repository README and RFAP-1's formalization appendix.

References

[1] N. J. Wildberger, *One dimensional metrical geometry*, arXiv:math/0701338, 2007.

[2] M. Beeler, R. W. Gosper, and R. Schroeppel, *HAKMEM*, MIT Artificial Intelligence Laboratory Memo AIM-239, 1972.

[3] J. M. Borwein, N. J. Calkin, S. B. Lindstrom, and A. Mattingly, Continued logarithms and associated continued fractions, *Experimental Mathematics* **26** (2017), no. 4, 412–429.

[4] J. M. Borwein, K. G. Hare, and J. G. Lynch, Generalized continued logarithms and related continued fractions, *Journal of Integer Sequences* **20** (2017), Article 17.5.7.

[5] M. J. Collins, Computing with continued logarithms, *arXiv:2602.07245*, 2026.

[6] M. Hockman, The geometry of Gaussian integer continued fractions, *Journal of Number Theory* **197** (2019), 145–167.

[7] A. Hurwitz, Über die Entwicklung complexer Grössen in Kettenbrüche, *Acta Mathematica* **11** (1887), 187–200.

[8] M. S. Kirsebom, Extreme value theory for Hurwitz complex continued fractions, *Journal of Theoretical Probability* **34** (2021), 2166–2197.

[9] G. González Robert, *Complex Continued Fractions*, Ph.D. thesis, Aarhus University, 2018.

[10] J. Shallit, Length of the continued logarithm algorithm on rational inputs, *Integers* **20** (2020), Paper A69.